

$$\begin{aligned}
\text{the likelihood: } L(\pi) &= \prod_{i=1}^N f(y_i) \\
&= \prod_{i=1}^N \pi^{y_i} (1 - \pi)^{1-y_i} \\
&= (\textit{could simplify as before if we wanted...})
\end{aligned}$$